

SEN381 Assignment 3 - 3 Person Work Split

Based on the assignment brief: each member owns one 15-mark research question, while Question 4 and the final document are completed collaboratively.

Person 1 – Quality Engineering (Q1)

- QA vs QC vs Verification vs Validation
- Risk-based testing and verification
- Compare three verification methods
- Explain strengths and limitations
- Automation and quality gates
- Create the Risk-to-Verification Evidence Map
- Answer the Q1 critical question

Person 2 – Security Engineering (Q2)

- Research threat modelling (e.g. STRIDE)
- Select four security concern areas
- Create threat → requirement → control → evidence chains
- Compare secure design, coding and dependency checks
- Evaluate two limitations of automated security tools and AI
- Complete the Threat-to-Control Traceability Table
- Answer the Q2 critical question

Person 3 – Production Readiness (Q3)

- Environment strategy (Dev/Test/Production)
- Secrets and configuration management
- Deployment and release process
- Rollback, backups and recovery
- Logs, metrics, monitoring and alerts
- Create the Production-Readiness Evidence Matrix
- Answer the Q3 critical question

Shared Responsibilities (Everyone)

Task	Responsibility
Q4 Research-to-M3 Engineering Map	Everyone
Cover page & Table of Contents	Everyone
Introduction & Conclusion	Everyone
Reference list (8+ sources)	Everyone
In-text citation check	Everyone

AI Research & Verification Record	Everyone (if used)
Proofreading & formatting	Everyone
Final Pull Request review	Everyone

Suggested GitHub Branches

- feature/Q1-quality
- feature/Q2-security
- feature/Q3-production
- Merge into develop via Pull Requests
- Final formatting PR before submission